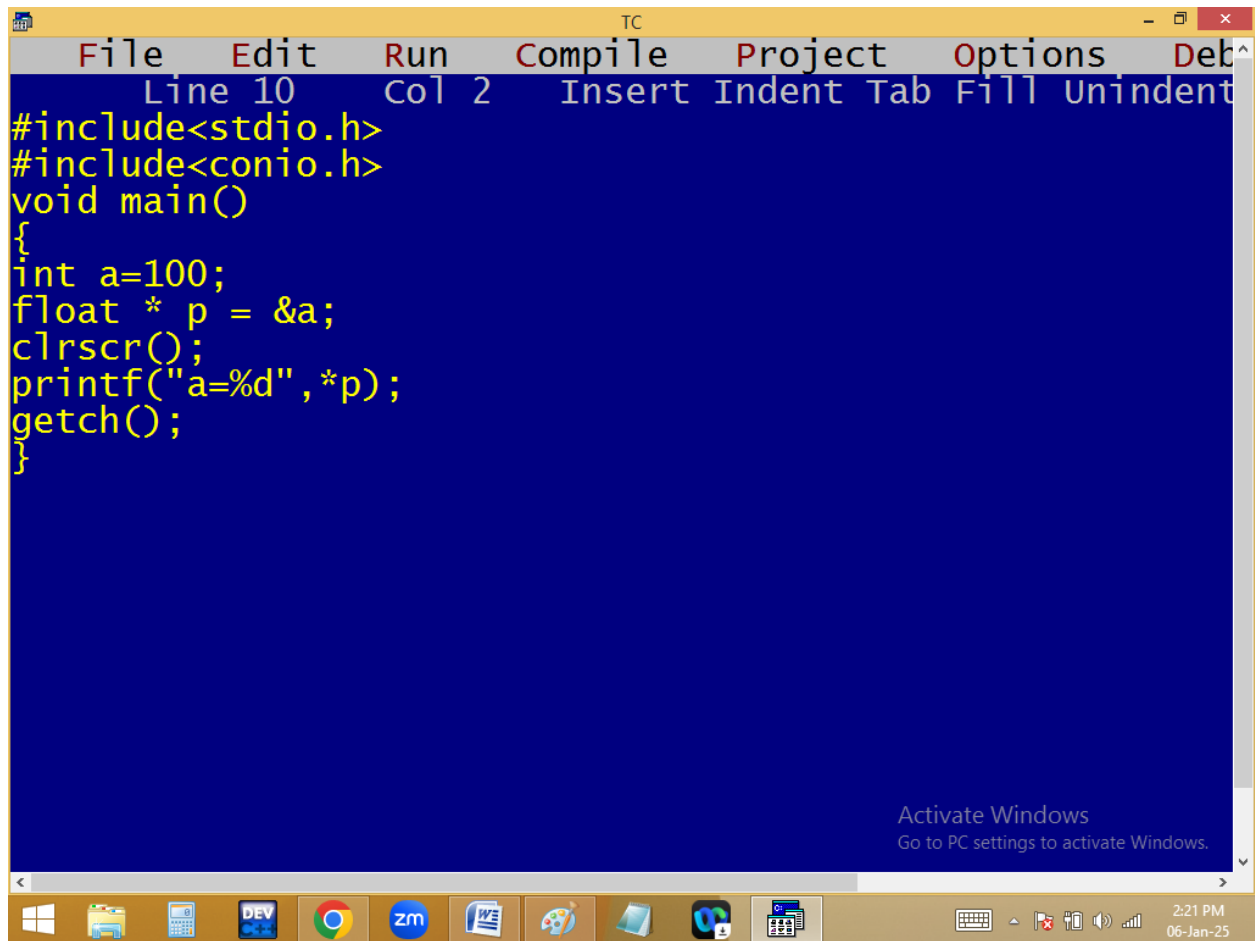


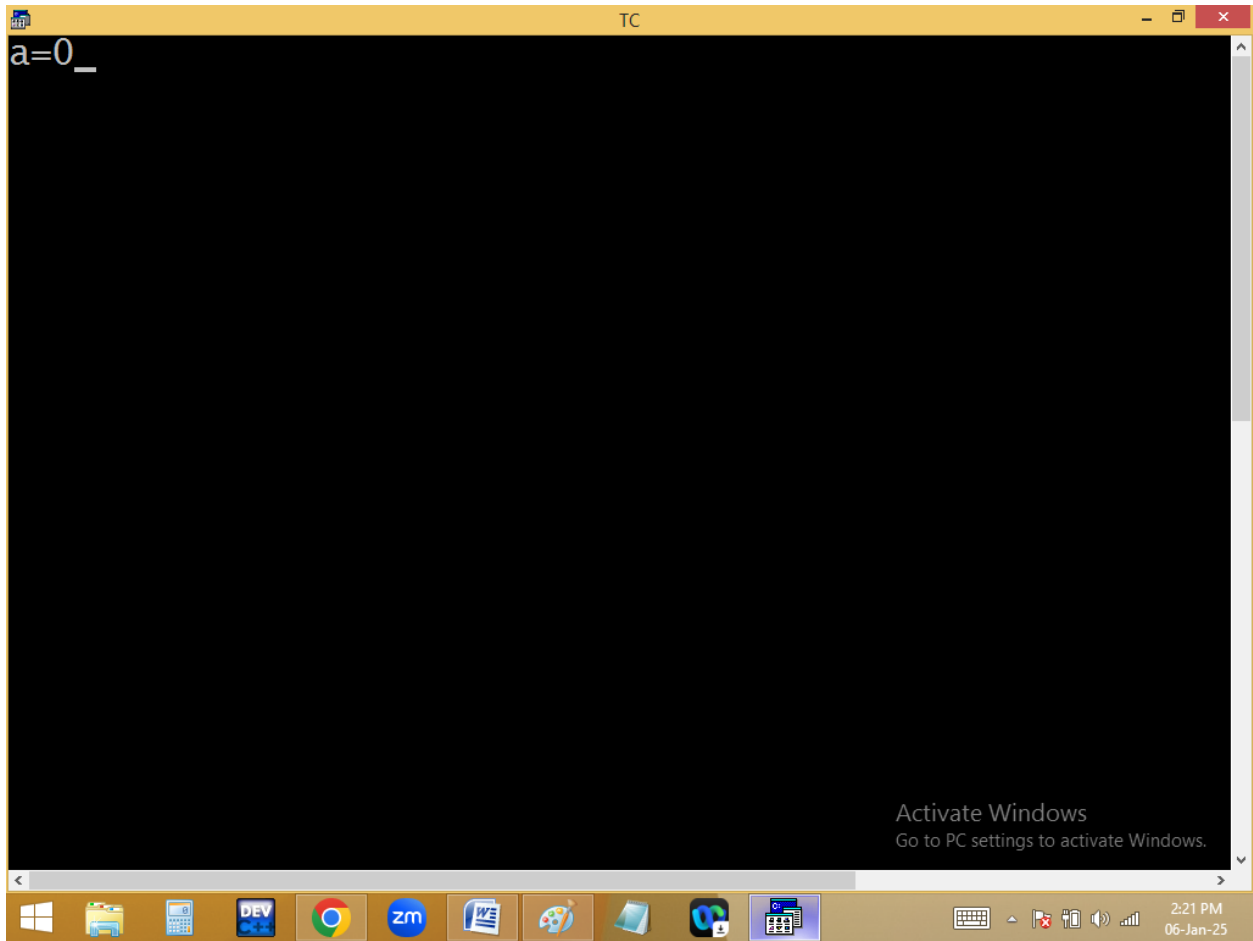
Pointer compatibility: Pointer can store only the same type of variable address and when different type address is given, it return garbage or runtime error.

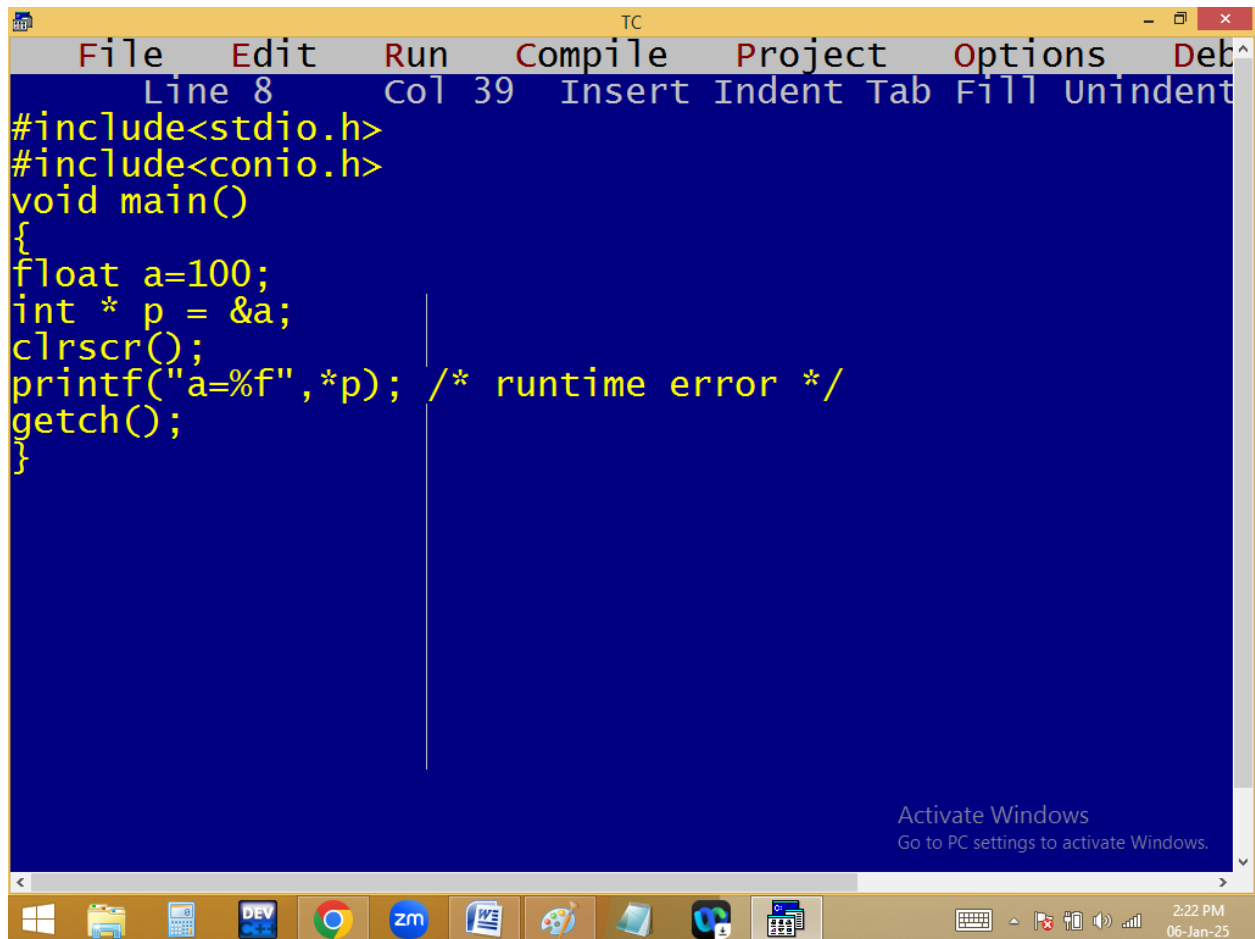


```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 2 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100;
float * p = &a;
clrscr();
printf("a=%d",*p);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

2:21 PM
06-Jan-25



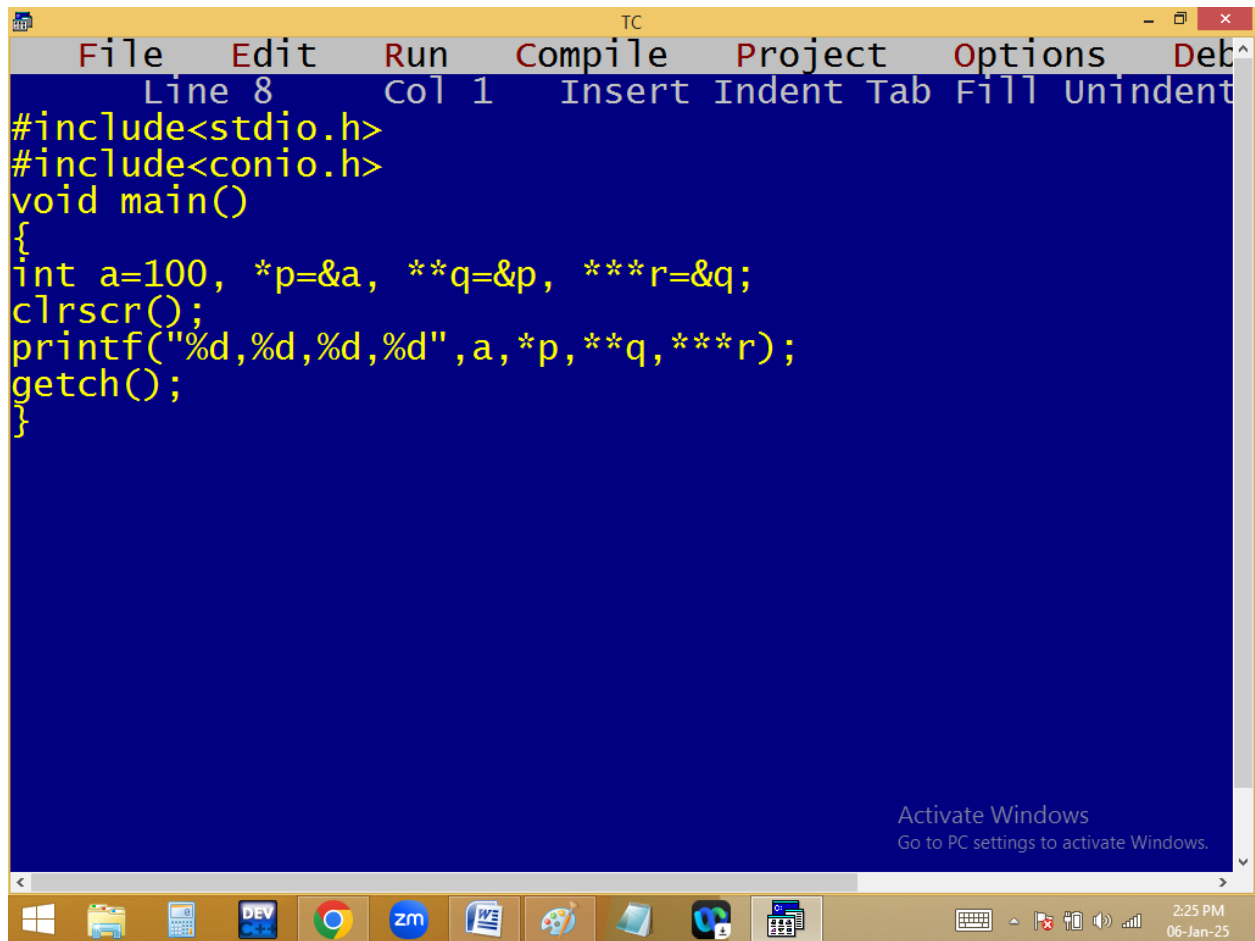


The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 8' and 'Col 39'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float a=100;
int * p = &a;
clrscr();
printf("a=%f",*p); /* runtime error */
getch();
}
```

The program attempts to print the value of a float variable 'a' through a pointer 'p' of type 'int'. This causes a runtime error because the pointer is not of the correct type to access the float data. The Windows taskbar at the bottom shows the time as 2:22 PM on 06-Jan-25.

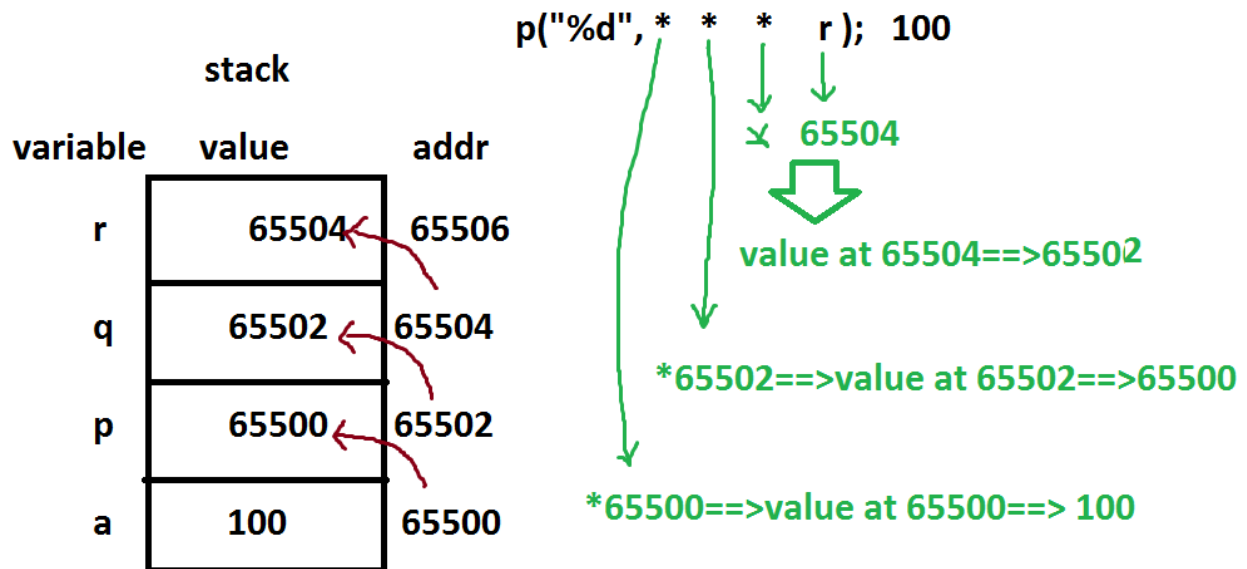
Double pointer / pointer to pointer: the pointer which stores the address of another pointer is called double pointer. They are used to handle dynamic multi dimensional arrays.



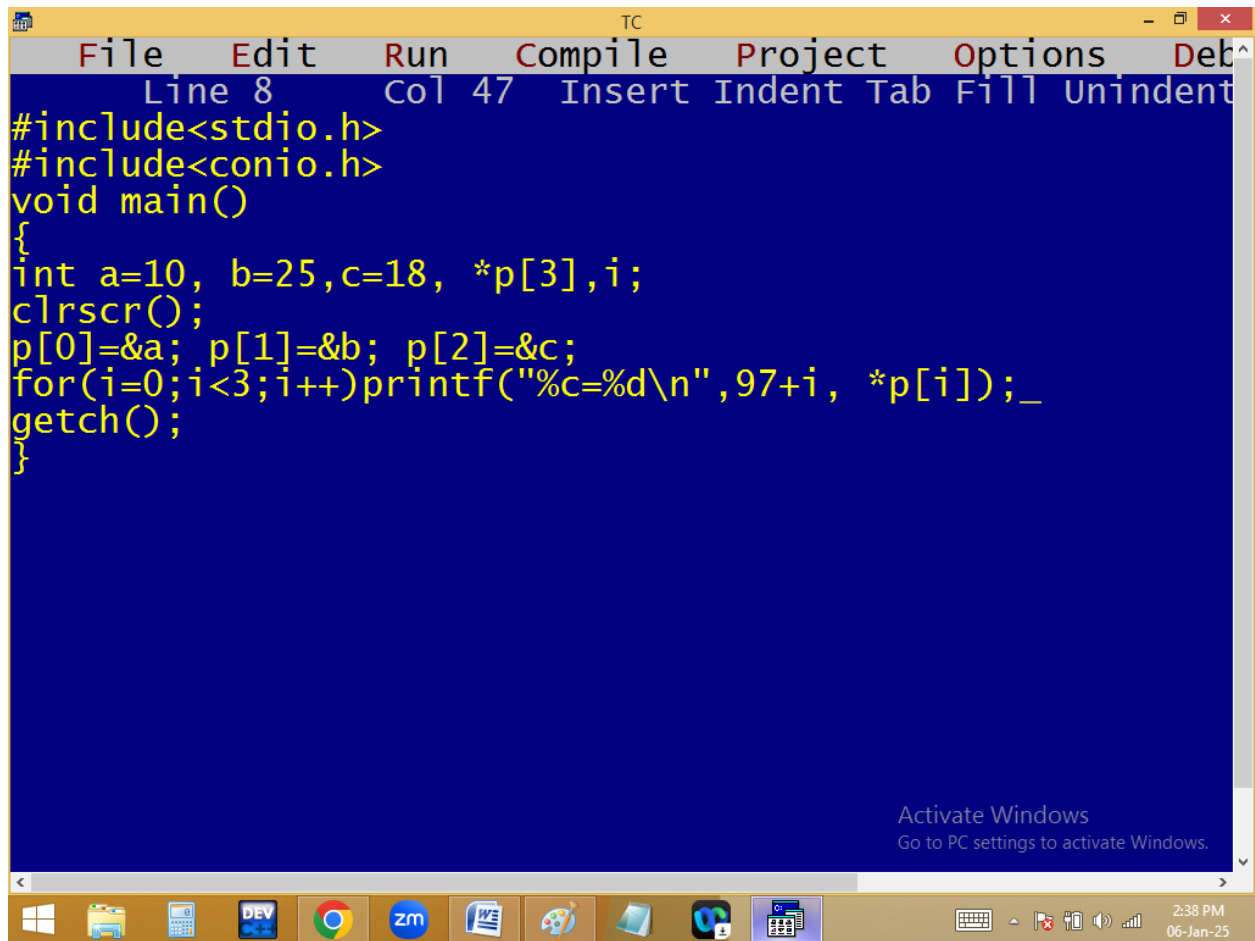
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 8", "Col 1", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p=&a, **q=&p, ***r=&q;
clrscr();
printf("%d,%d,%d,%d",a,*p,**q,***r);
getch();
}
```

In the bottom right corner of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, File History, and OneDrive. The system tray on the right shows the date and time as "2:25 PM" and "06-Jan-25".



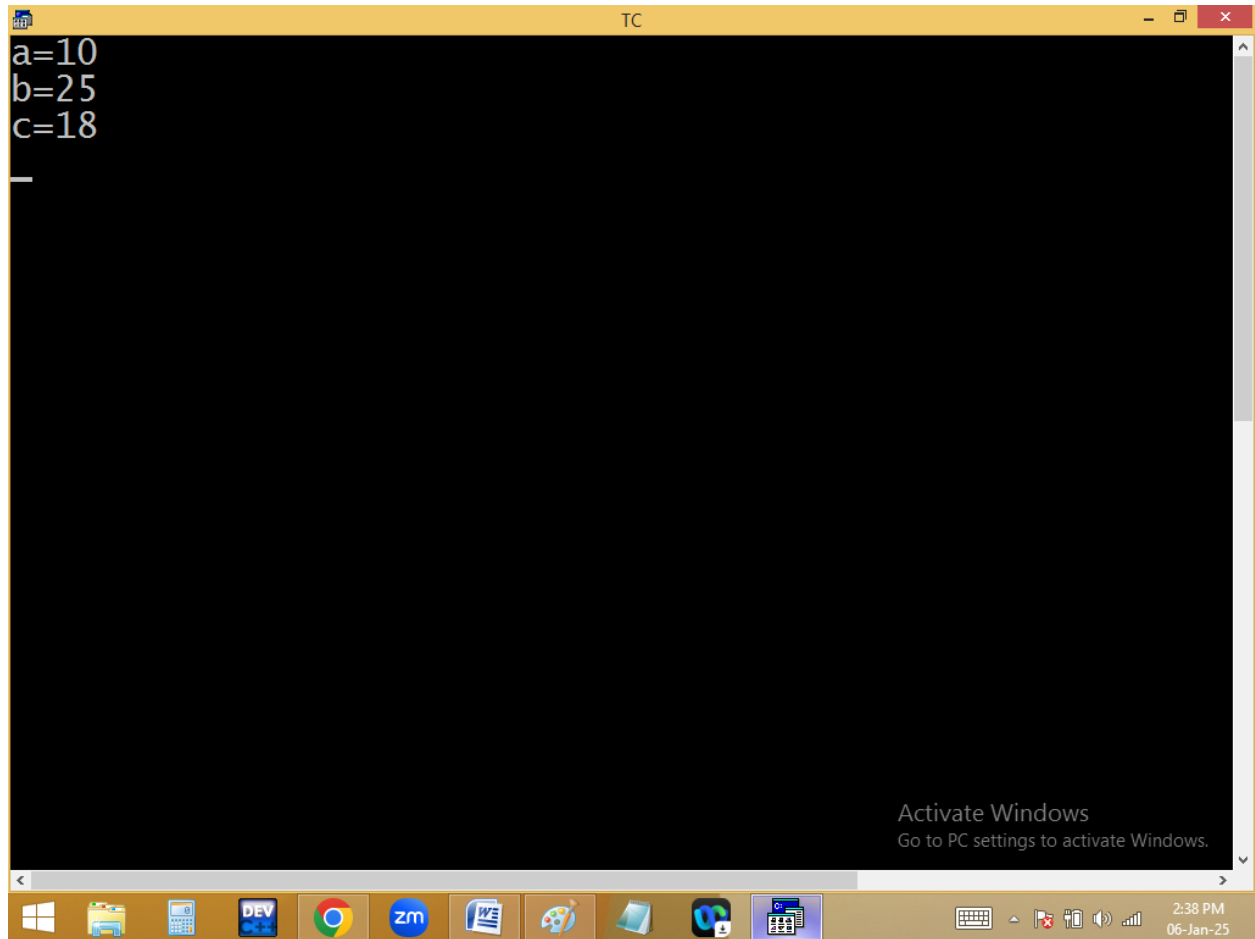
Array of Pointer: Like general variables we can declare pointer using array. Due to this one pointer variable can store multiple variable address. It is used to handle dynamic multi dimensional arrays.

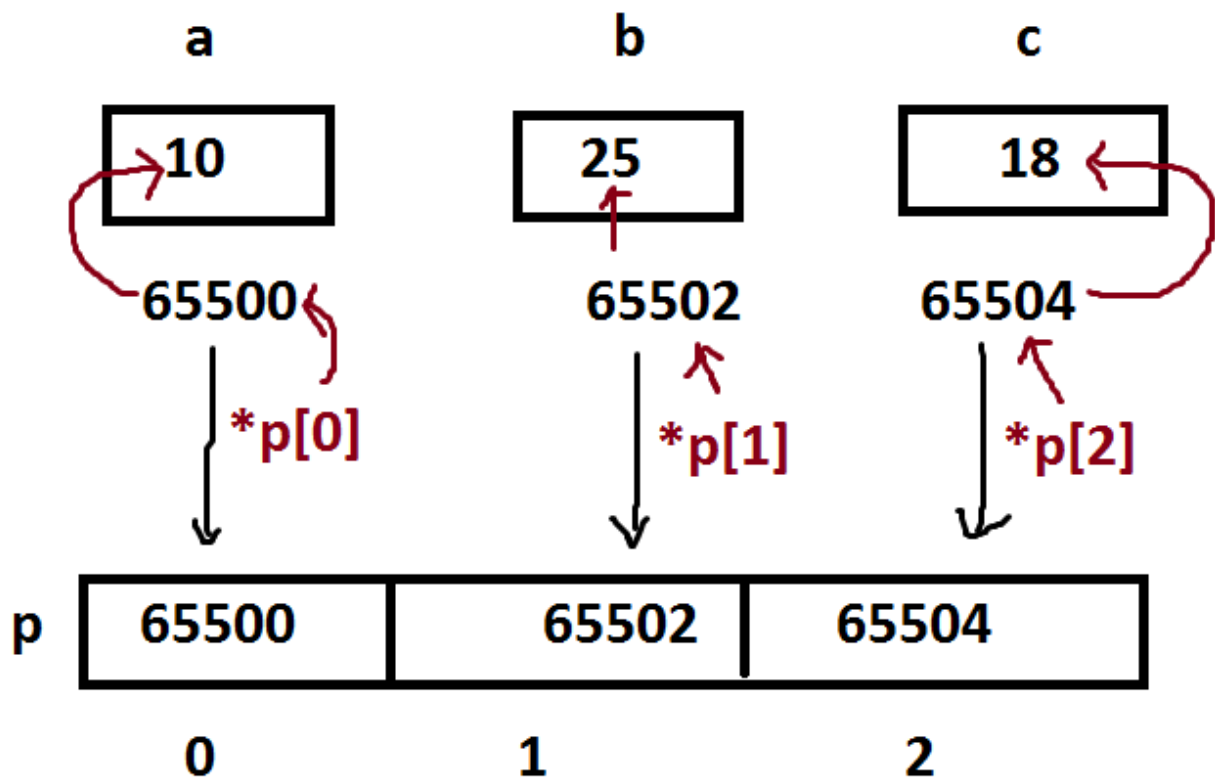


```
TC
File Edit Run Compile Project Options Deb
Line 8 Col 47 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=25,c=18, *p[3],i;
clrscr();
p[0]=&a; p[1]=&b; p[2]=&c;
for(i=0;i<3;i++)printf("%c=%d\n",97+i, *p[i]);_
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

2:38 PM
06-Jan-25





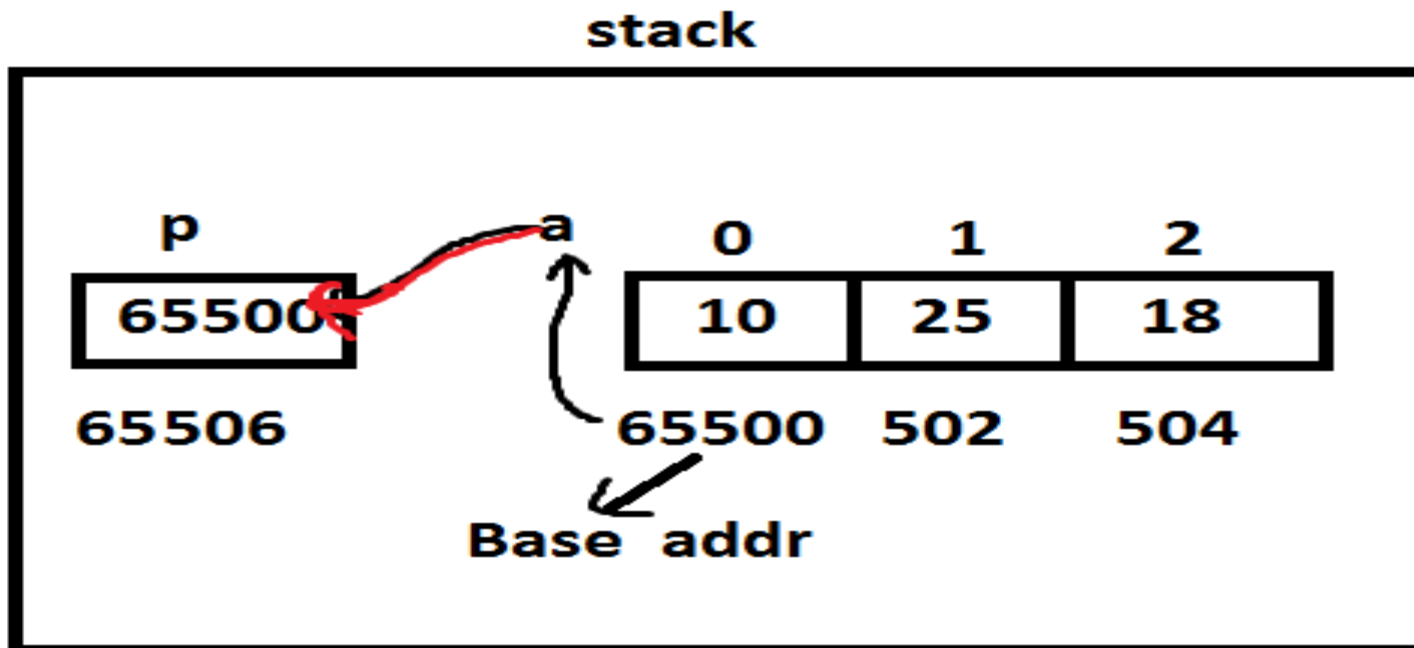
Pointer to array:

Array is implicit pointer. Due to this it holds the base cell addr [0 cell addr] implicitly. By assigning the array name or 0 cell addr to the pointer, we can handle array elements using the following syntax.

`*(ptrvariable + offset/index * sizeof(variable));`

Eg:

```
int a[3]={10, 25, 18}, *p, i;  
p = a ; or p = &a[0]; or p = &a;
```



```
for(i=0;i<3;i++)  
printf("%4d", *(p+i));
```

Here $*(p+i)$ meaning is:

p is 65500

1. $*(p+0*2) \rightarrow *65500 \rightarrow \text{value at } 65500 \rightarrow 10$
2. $*(p+1*2) \rightarrow *65502 \rightarrow \text{value at } 65502 \rightarrow 25$
3. $*(p+2*2) \rightarrow *65504 \rightarrow \text{value at } 65504 \rightarrow 18$

Note: Here 2 is int size.

Eg:

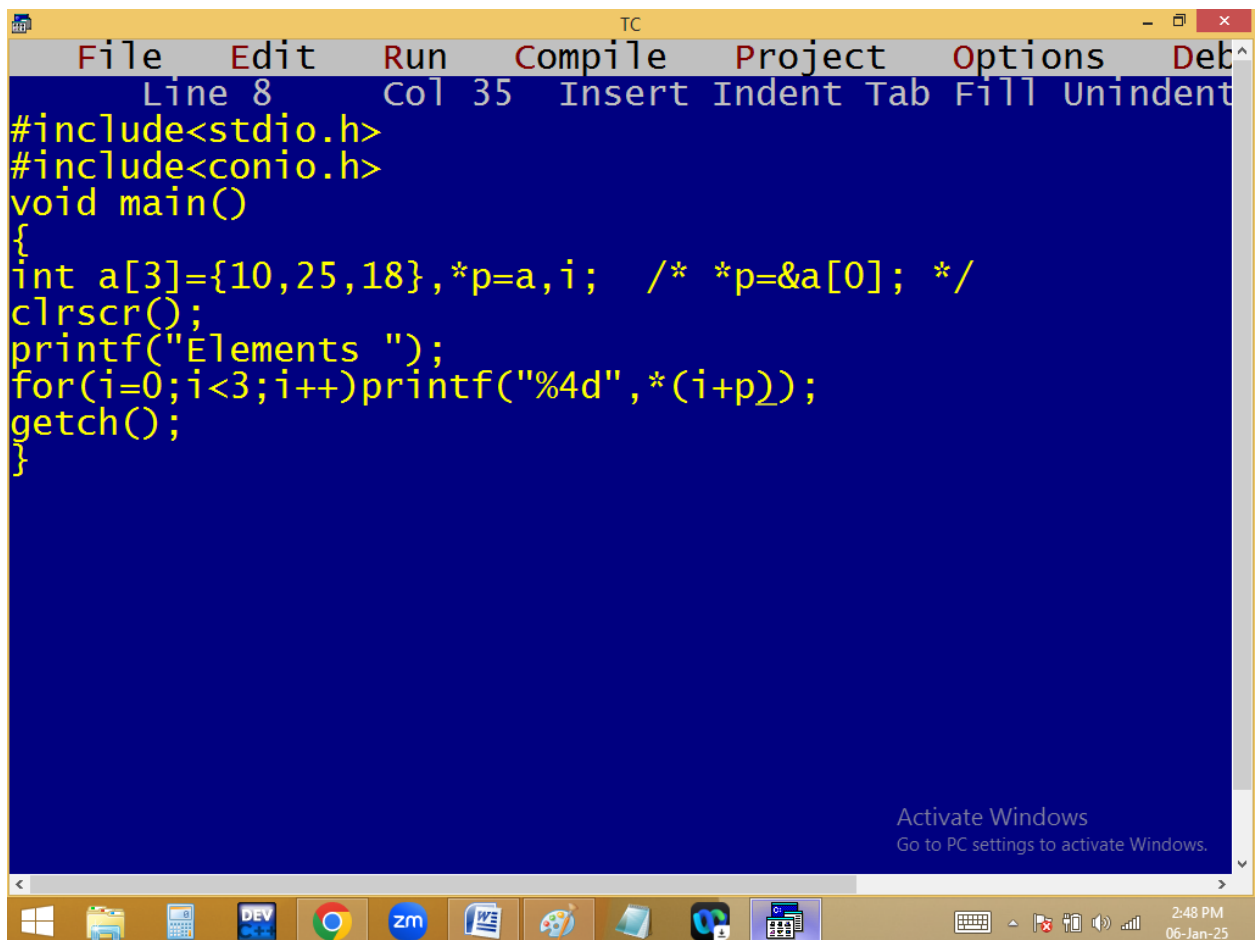
```
#include<stdio.h>  
#include<conio.h>  
void main()  
{  
int *p, a[3]={10,25,18}, i;  
clrscr();  
p = a; /* p=&a[0]; or p=&a; */
```

```
printf("Elements are: ");  
for(i=0;i<3;i++)  
printf("%4d",*(p+i));  
getch();  
}
```

Output: Elements are: 10 25 18

Note: We can access array elements using array / pointer in following ways.

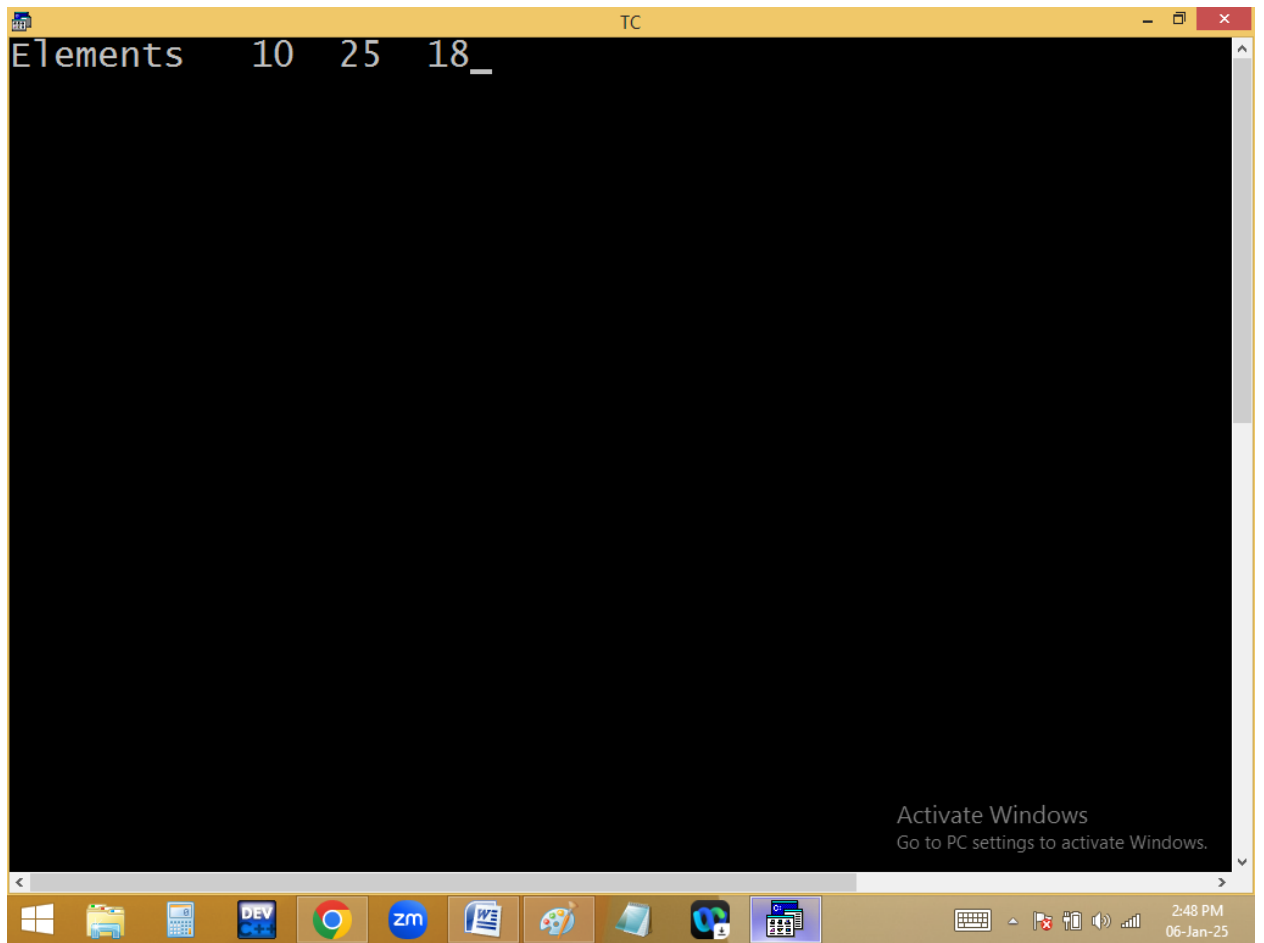
a[i] / i[a] / p[i] / i[p] / *(p+i) / *(a+i) / *(i+p) / *(i+a)

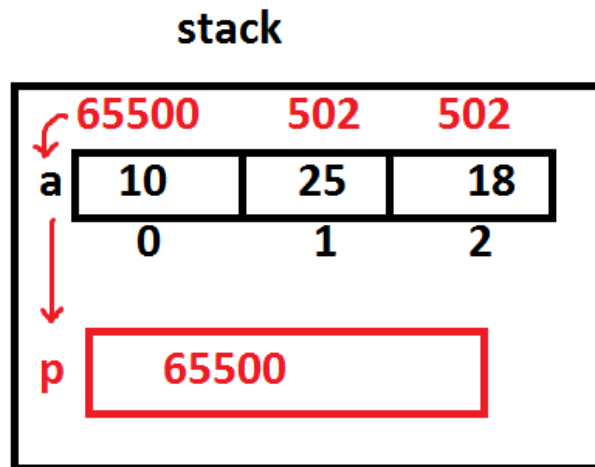


The screenshot shows a Turbo C++ IDE window titled 'TC'. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 8 Col 35' and lists keyboard shortcuts: Insert, Indent, Tab, Fill, and Unindent. The code in the editor is as follows:

```
#include<stdio.h>  
#include<conio.h>  
void main()  
{  
int a[3]={10,25,18},*p=a,i; /* *p=&a[0]; */  
clrscr();  
printf("Elements ");  
for(i=0;i<3;i++)printf("%4d",*(i+p));  
getch();  
}
```

At the bottom of the window, there is a message: 'Activate Windows Go to PC settings to activate Windows.' The Windows taskbar is visible at the very bottom, showing icons for File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, and other applications. The system clock shows 2:48 PM on 06-Jan-25.





```
for( i=0; i<3; i++ ) p( *(p+i) );
```

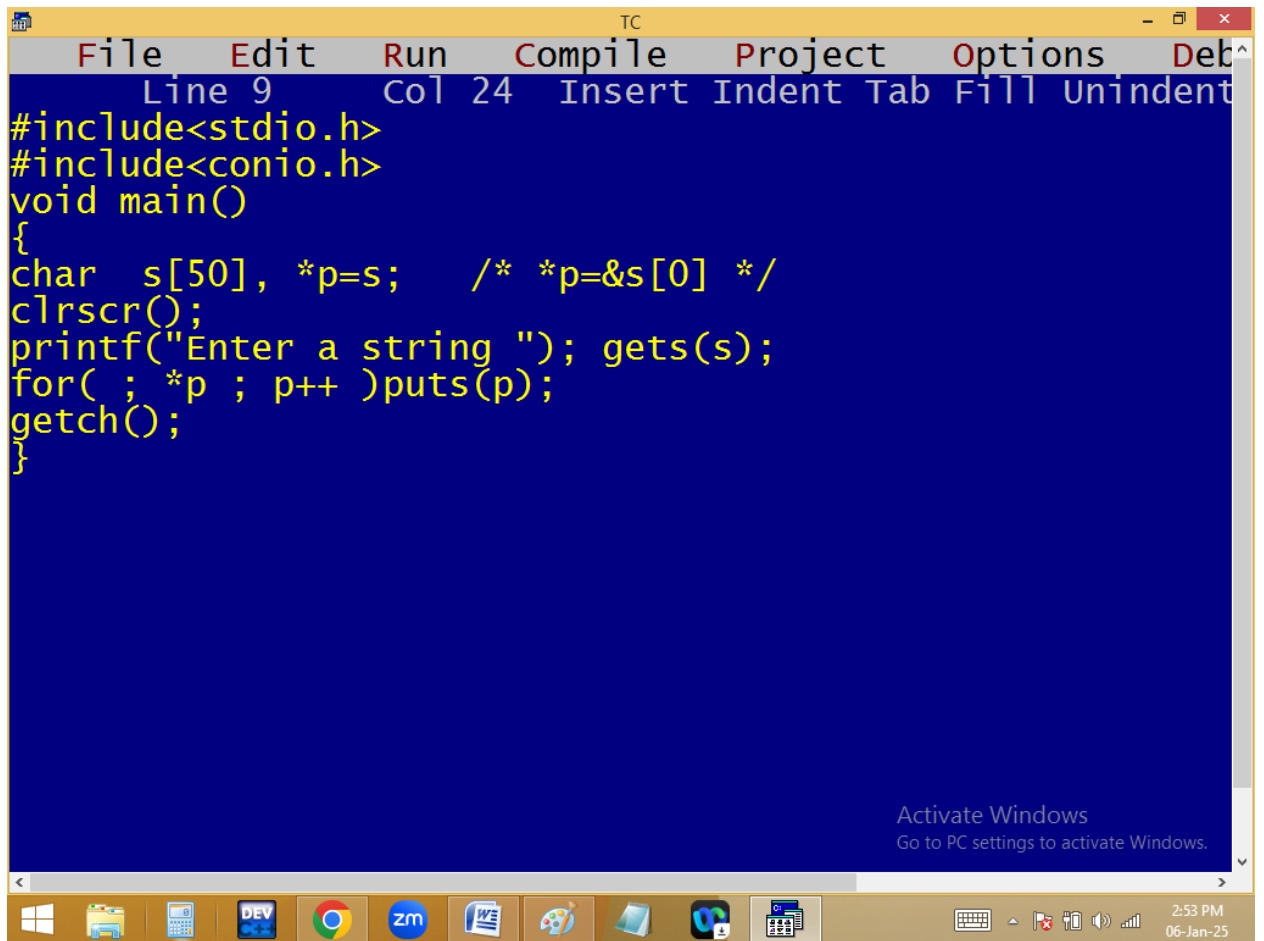


$*65500+0*2=*65500 \Rightarrow \text{value at } 65500=10$

$*65500+1*2=*65502 \Rightarrow \text{value at } 65503=25$

$*65500+2*2=*65504 \Rightarrow \text{value at } 65504=18$

Pointer to string:



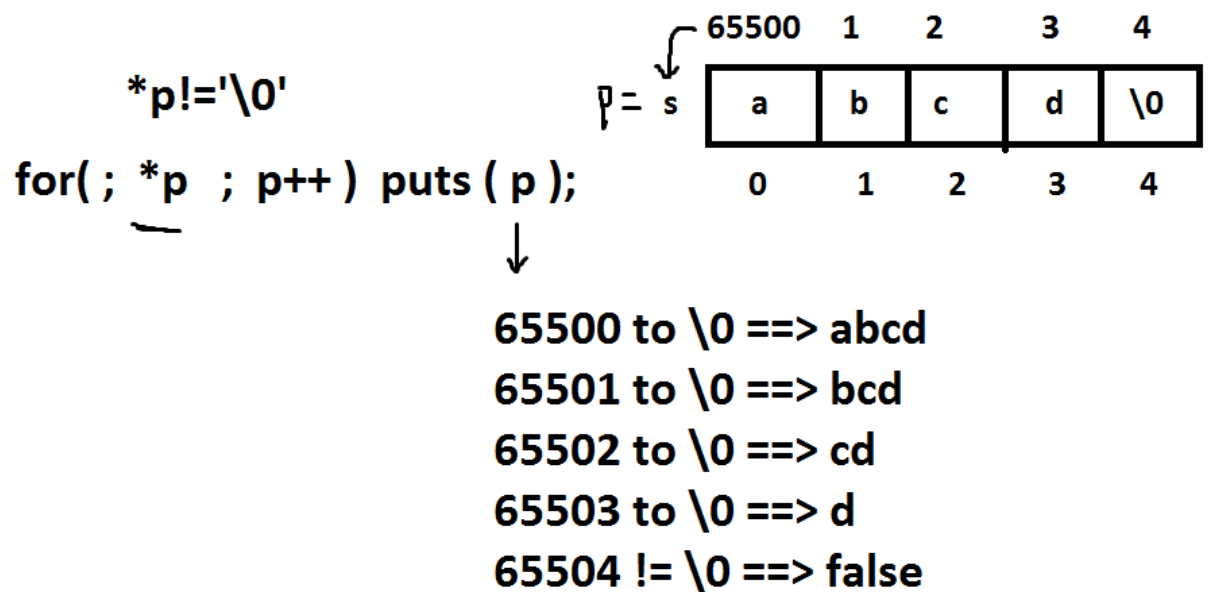
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the bottom of the window displays "Line 9", "Col 24", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char  s[50], *p=s;    /* *p=&s[0] */
clrscr();
printf("Enter a string "); gets(s);
for( ; *p ; p++ )puts(p);
getch();
}
```

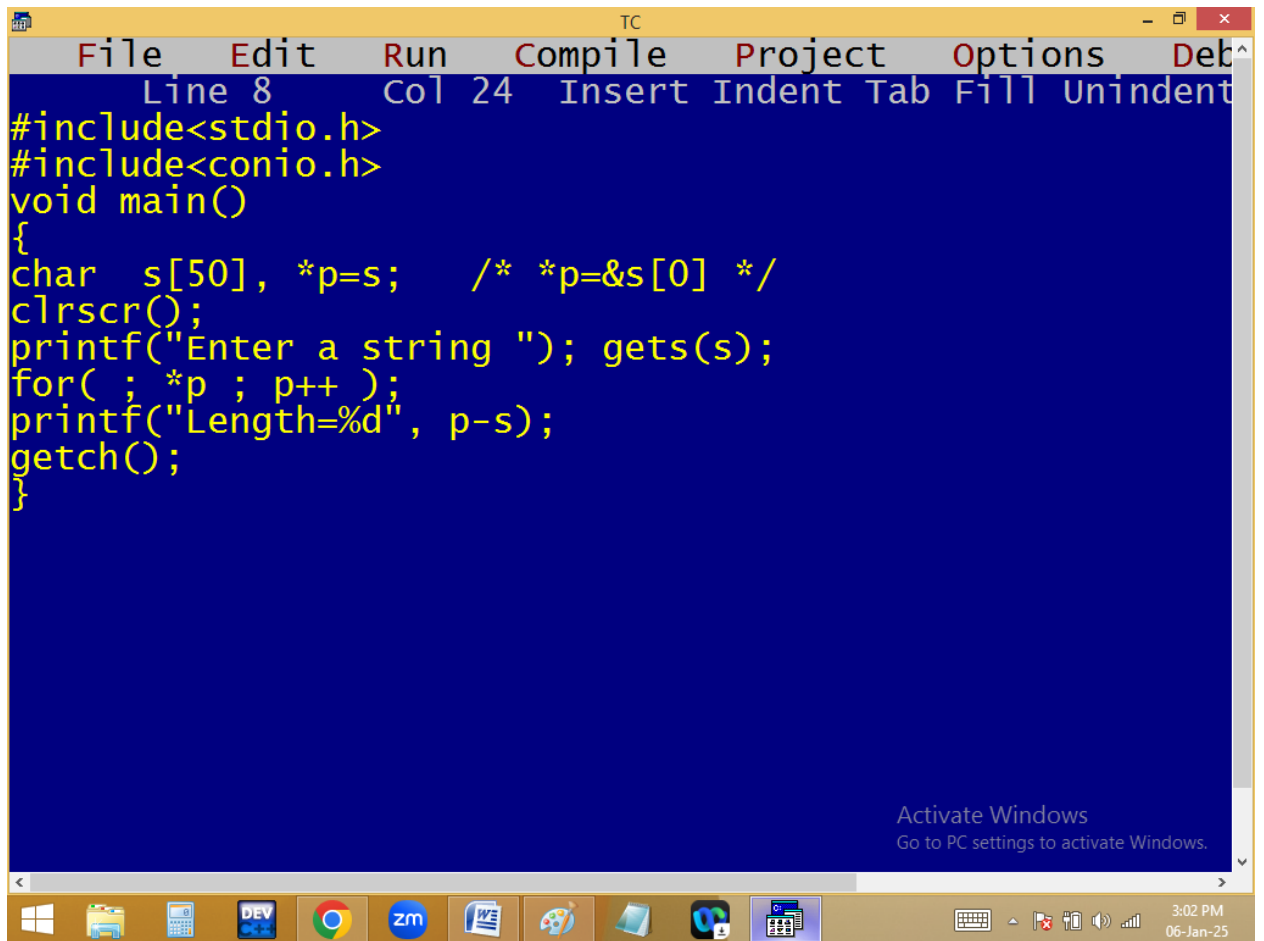
In the bottom right corner of the IDE window, there is a message: "Activate Windows. Go to PC settings to activate Windows." The Windows taskbar is visible at the very bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and a folder icon. The system tray on the right shows the date and time as "2:53 PM 06-Jan-25" along with icons for keyboard, network, and volume.

```
TC
Enter a string Bharathi
Bharathi
harathi
arathi
rathi
athi
thi
hi
i
```

Activate Windows
Go to PC settings to activate Windows.



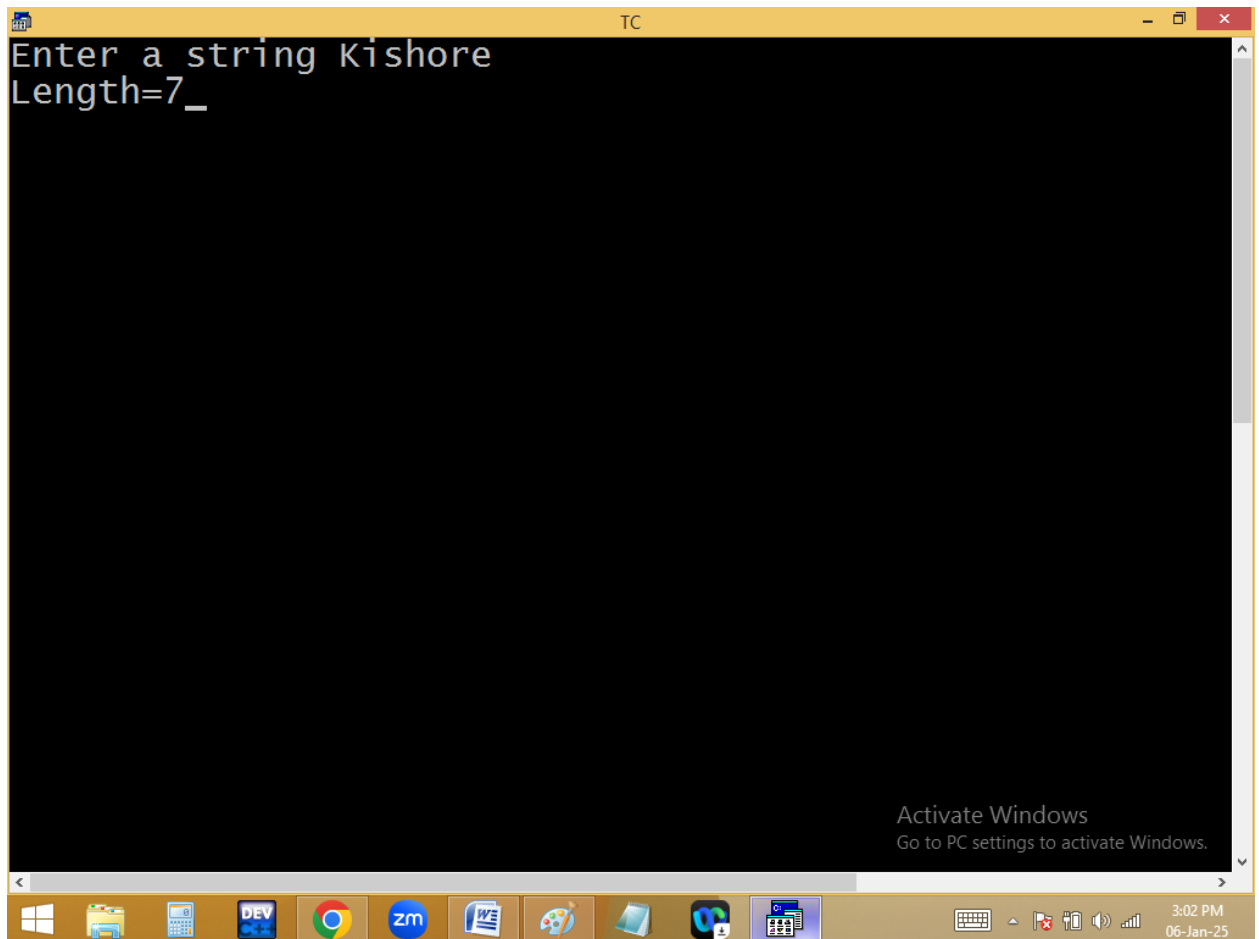
Find a string length using pointer only:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 8", "Col 24", and a list of editing actions: Insert, Indent, Tab, Fill, and Unindent. The main editor area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char  s[50], *p=s;    /* *p=&s[0] */
clrscr();
printf("Enter a string "); gets(s);
for( ; *p ; p++ );
printf("Length=%d", p-s);
getch();
}
```

In the bottom right corner of the editor area, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom of the screen, showing various application icons including File Explorer, Calculator, Dev-C++, Google Chrome, Zoom, Word, and others. The system clock in the bottom right corner indicates the time is 3:02 PM on 06-Jan-25.



The screenshot shows a Turbo C++ (TC) window with a black background. The text "Enter a string Kishore" and "Length=7_" is displayed in white. The window title bar says "TC". The Windows taskbar is visible at the bottom with various icons and a system clock showing 3:02 PM on 06-Jan-25.

```
Enter a string Kishore
Length=7_
```

Pointer arithmetic: Like general variable we can do =, ==, +, -, ++, -- on pointers also. But we can't do *, %, / on pointers.

TC

File Edit Run Compile Project Options Deb^

Line 10 Col 1 Insert Indent Tab Fill Unindent

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p=&a;
clrscr();
p*=2;
p%=2;
p/=2;
getch();
}
```

Compiling

Main file: 2PM.C
Compiling: EDITOR → 2PM.C

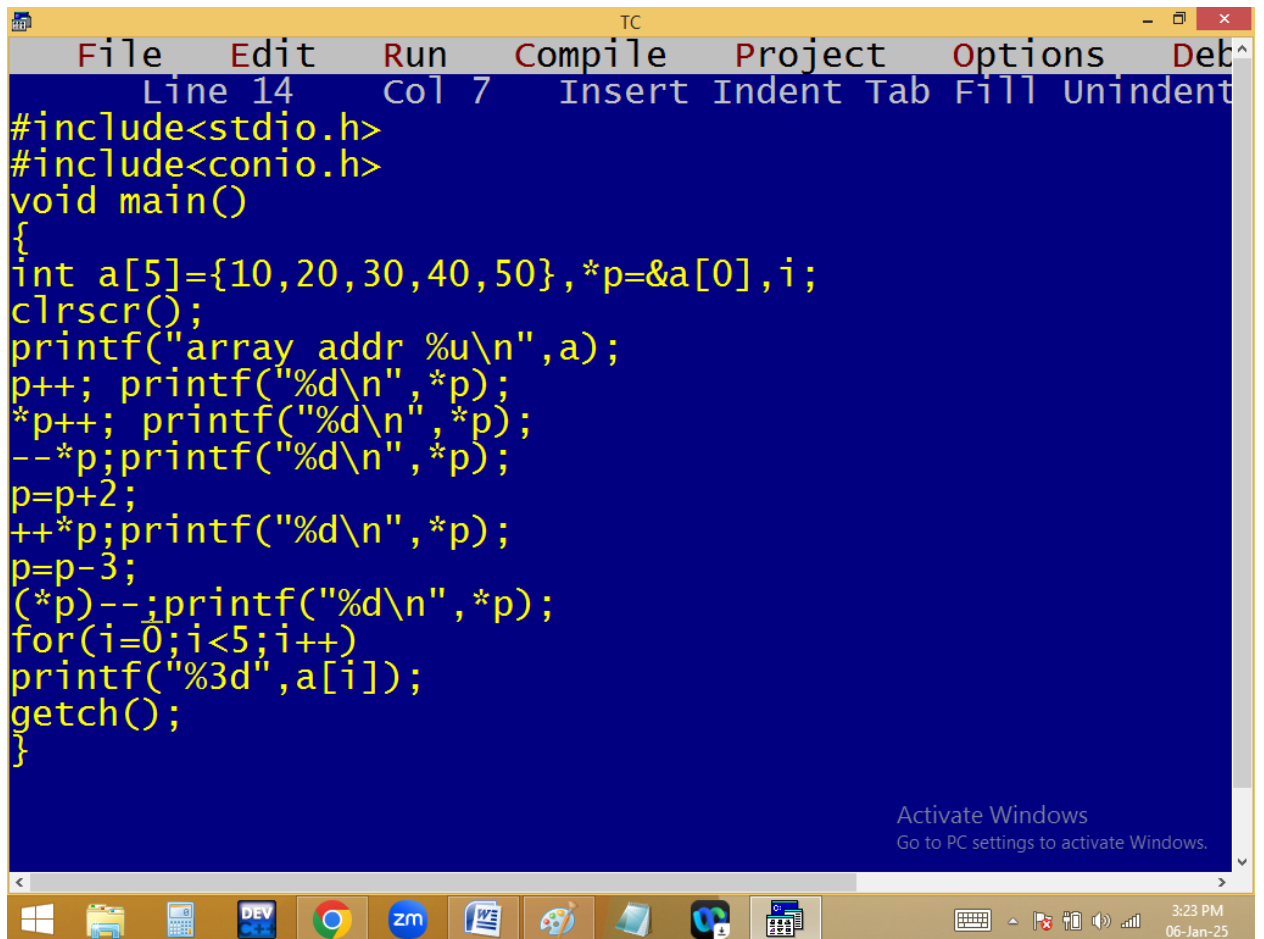
	Total	File
Lines compiled:	321	321
Warnings:	1	1
Errors:	3	3

Available memory: 250K

Errors : Press any key

Activate Windows
Go to PC settings to activate Windows.

3:06 PM
06-Jan-25



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the bottom of the window displays "Line 14", "Col 7", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent".

The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a[5]={10,20,30,40,50},*p=&a[0],i;
    clrscr();
    printf("array addr %u\n",a);
    p++; printf("%d\n",*p);
    *p++; printf("%d\n",*p);
    --*p;printf("%d\n",*p);
    p=p+2;
    ++*p;printf("%d\n",*p);
    p=p-3;
    (*p)--;printf("%d\n",*p);
    for(i=0;i<5;i++)
        printf("%3d",a[i]);
    getch();
}
```

In the bottom right corner of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows.".

The Windows taskbar is visible at the very bottom of the image, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, File History, OneDrive, and Task View. The system tray on the right shows the date and time as "3:23 PM 06-Jan-25".

```

array addr 65494
20
30
29
51
19
10 19 29 40 51_

```

Activate Windows
Go to PC settings to activate Windows.

$p++ \Rightarrow 65494 + 1 \times 2 = 65496$
 $p(*p) \Rightarrow \text{value at } 65496 \Rightarrow 20 \checkmark$

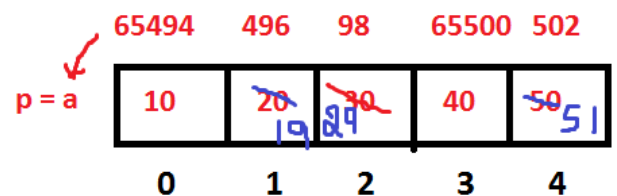
$*p++ \Rightarrow 65496 + 1 \times 2 = 65498$
 $p(*p) \Rightarrow \text{value at } 65498 \Rightarrow 30 \checkmark$

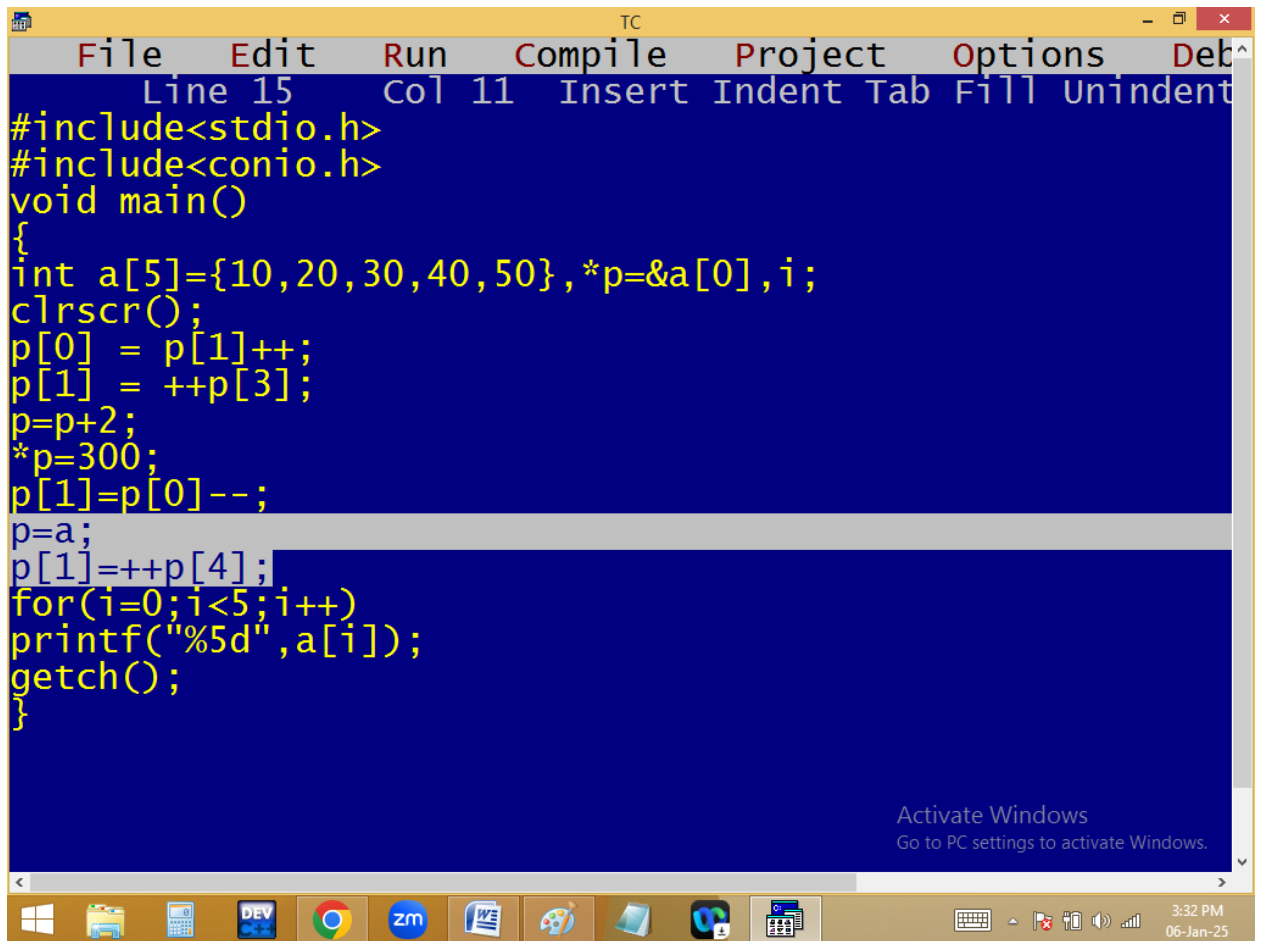
$--*p \Rightarrow \text{value at } 65498-- \Rightarrow --30 \Rightarrow 29$

$p(*p) \Rightarrow \text{value at } 65498 \Rightarrow 29 \checkmark$

$p=p+2 \Rightarrow 65498 + 2 \times 2 = 65502$
 $++*p \Rightarrow \text{value at } 65502++ \Rightarrow ++50 \Rightarrow 51$
 $p(*p) \Rightarrow \text{value at } 65502 \Rightarrow 51 \checkmark$

$p=p-3 \Rightarrow 65502 - 3 \times 2 = 65496$
 $(*p)-- \Rightarrow \text{value at } 65496-- \Rightarrow 20-- \Rightarrow 19$
 $p(*p) \Rightarrow \text{value at } 65496 \Rightarrow 19 \checkmark$

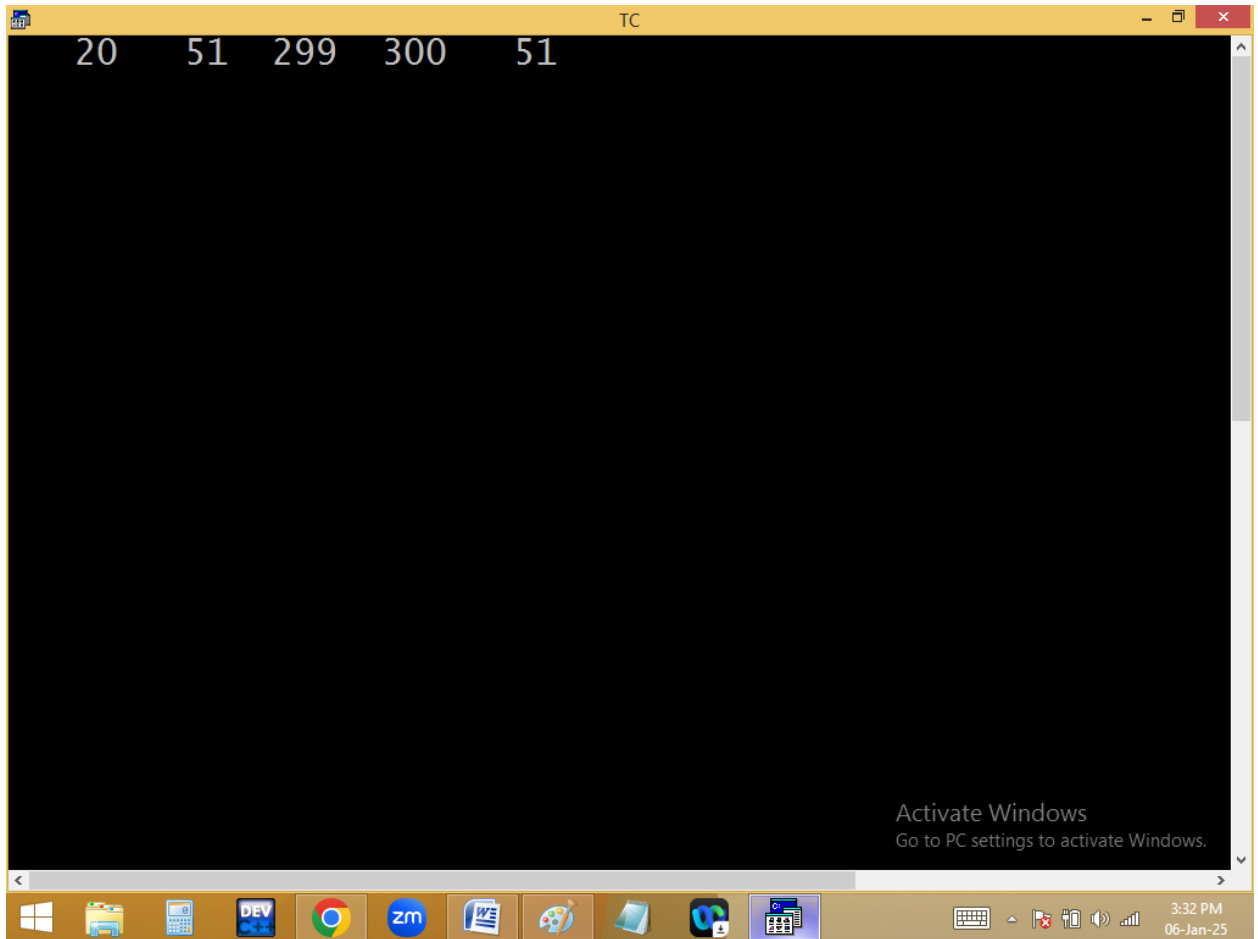




```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 11 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a[5]={10,20,30,40,50},*p=&a[0],i;
clrscr();
p[0] = p[1]++;
p[1] = ++p[3];
p=p+2;
*p=300;
p[1]=p[0]--;
p=a;
p[1]=++p[4];
for(i=0;i<5;i++)
printf("%5d",a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

3:32 PM
06-Jan-25



$p[0] = p[1]++;$ $\Rightarrow p[0]=p[1] \Rightarrow p[1]++ \Rightarrow 21$

$p[1] = ++p[3] \Rightarrow ++p[3] \Rightarrow ++40 \Rightarrow 41$

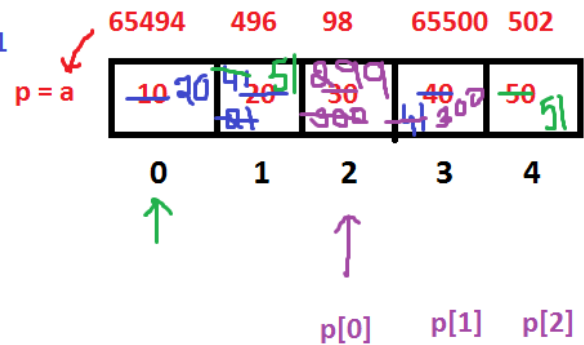
$p[1] = p[3]$

$p=p+2 \Rightarrow 65494+2*2=65498$

$*p = 300 \Rightarrow 65498 = 300$

$p[1] = p[0] -- \Rightarrow p[1] = p[0] \Rightarrow 41 = 300$

$p[0]-- \Rightarrow 300 -- \Rightarrow 299$



$p=a \Rightarrow 65494$

$p[1]= ++p[4] \Rightarrow ++p[4] \Rightarrow ++50 \Rightarrow 51$

$p[1] = p[4] \Rightarrow 41 = 51$